BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 310 (July 2023)

Communications Laboratory

Final Project Report

Section: A2 Group: 05

Pulse Code Modulation (PCM)

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Signature of Instructor: ______________________________

Academic Honesty Statement:

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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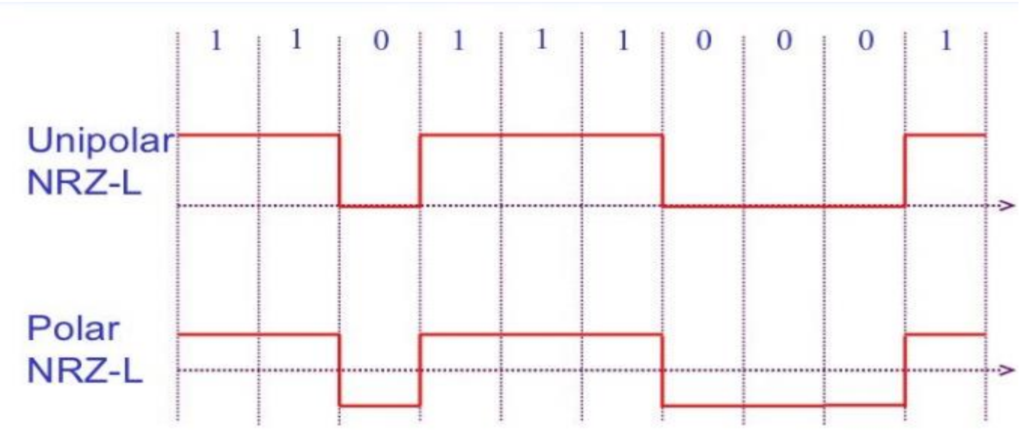
1 Abstract

This project aims to implement Pulse Code Modulation (PCM), an important technique in digital communication systems for transforming analog signals into a digital format. PCM achieves this by sampling the analog signal at regular intervals and then quantizing each sample into discrete values. Through our experiment, we explored the theoretical foundations of PCM, including its sampling rate, quantization levels, and encoding techniques.Through analysis and experimentation, this project aims to provide a comprehensive understanding of PCM principles and its wide-ranging applications in modern telecommunications and digital audio technology.

2 Introduction

Pulse Code Modulation (PCM) is a fundamental method in digital communication and signal processing that has transformed the transmission and storage of analog signals. PCM converts analog signals into digital format by undergoing three key steps: Sampling, Quantization, and Encoding. Initially, the continuous analog signal is sampled at regular intervals, capturing its values over time. These samples are then quantized, where the amplitude values are discretized into distinct levels. Finally, the quantized samples are encoded into binary form meaning that there are only two possible states, represented by logic 1 (high) and logic 0 (low). Our goal is to design and implement a three-bit PCM (Pulse Code Modulation) encoding model. This method offers several advantages, including resistance to noise and distortion during transmission, compatibility with digital communication standards, and flexibility in signal processing.

For line coding NRZ-L is implemented. After quantization, each sample is encoded into a binary PCM code word. In NRZ-L, the binary codeword maintains the same voltage level (high or low) for the duration of the bit period. A '1' bit might be represented by a high voltage level, while a '0' bit is represented by a low voltage level. The voltage level remains constant until the next sample. Two types of NRZ-L are available unipolar and polar.



NRZ-L PCM offers simplicity and ease of implementation. However, it has limitations such as susceptibility to baseline wander (long strings of 0s or 1s) and lack of built-in clock recovery. These limitations can affect synchronization and signal integrity, especially in long-distance communication systems.

3 Design

3.1 Problem Formulation

3.1.1 Identification of Scope

The scope of the project on Pulse Code Modulation (PCM) encompasses a comprehensive exploration and implementation of the steps involved in digitizing analog signals, with a focus on sampling, quantization, encoding, and shift registering.

The project will investigate various sampling techniques, including natural sampling and flat-top sampling, to capture analog signals effectively. It will also analyze quantization techniques such as uniform quantization to determine the optimal approach for preserving signal quality. This project also analyzes the method how a encoder works and build an encoder from ground up using different logic gates. Implementation of shift registers will be carried out to facilitate the sequential transmission of PCM-encoded data. Investigation of shift register architectures and configurations will be conducted to optimize data serialization and synchronization.

Overall, the project's scope includes theoretical analysis, simulation, and practical implementation of PCM techniques, with an emphasis on optimizing signal quality, data efficiency, and transmission reliability through the integration of sampling, quantization, encoding, and shift registering steps.

3.1.2 Formulation of Problem

The objective of this project is to investigate and implement Pulse Code Modulation (PCM) as a method for digitizing analog signals. The project will proceed through several key steps: sampling, quantization, encoding, and finally, shift registering.

Sampling: The first step involves sampling the analog signal at regular intervals to capture its amplitude at discrete points in time. The sampling frequency must be carefully chosen to adequately represent the original signal while minimizing data redundancy and conserving bandwidth.

Quantization: Following sampling, the analog samples are quantized by assigning each sample to a specific discrete value within a predetermined range. This step involves determining the number of quantization levels and the corresponding bit depth required to accurately represent the analog signal while minimizing quantization errors.

Encoding: Once the analog samples are quantized, they are encoded into digital format for transmission or storage. This involves assigning a binary code to each quantized sample.

Shift Registering: Finally, the encoded digital data is processed through shift registers for efficient transmission or storage. Shift registers facilitate the sequential transfer of data bits, allowing for synchronization and serialization of the PCM data stream.

3.1.3 Analysis

3.2 Design Method

For the Pulse Code Modulation (PCM) project we used Proteus Simulation Software for simulation and implemented the circuit in the breadboard for physical implementation, the design method involves:

1. We first begin with requirement of system specifications and input analog signal. We are working with sinusoidal signal but with a little tweaking other signals can also be used.

2. We then designed and simulated each component individually in Proteus, including the Sampler, 3-bit Quantizer, Encoder, and Shift Register.

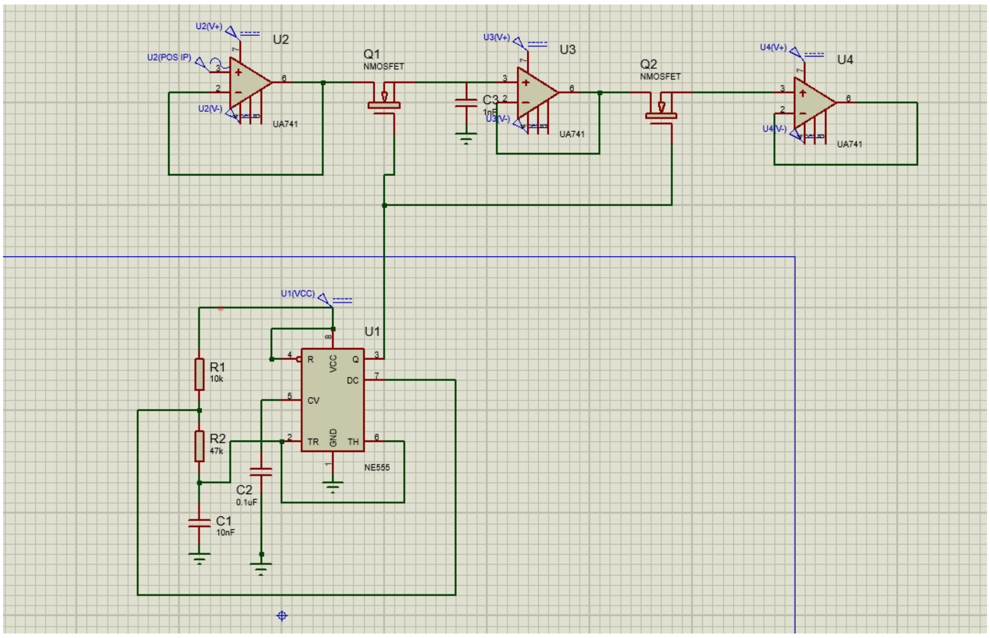
3. We then implement the simulated circuits on a breadboard, ensuring accurate replication of the designs and same values of resistances, capacitances and other elements.

4. We then integrated all components into a complete PCM system on the breadboard.

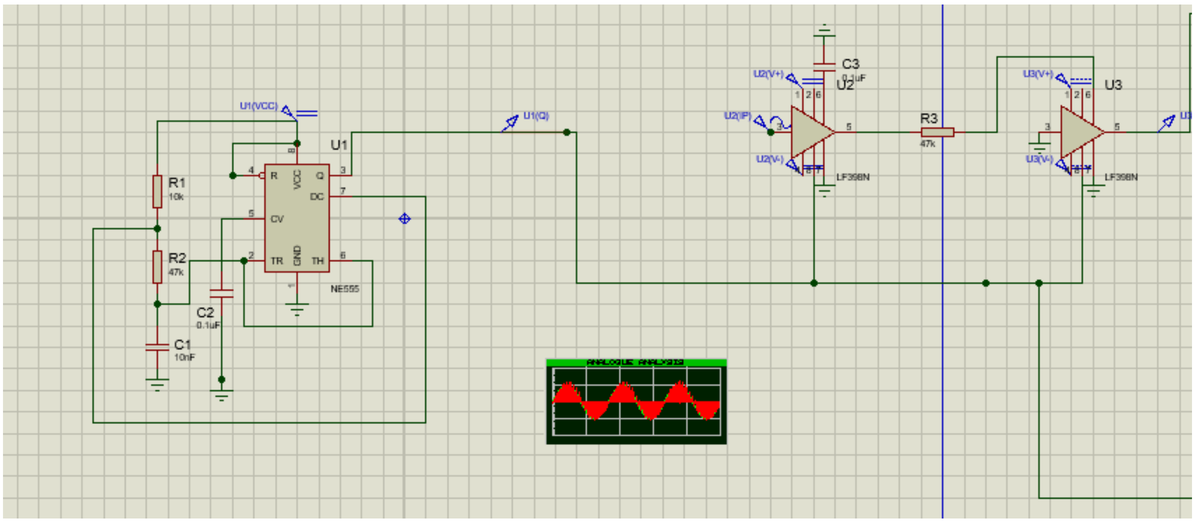
5. We then tried to verify the hardware implementation through practical testing, comparing results with simulations. In this step we faced a lot of problems regarding different biasing and outputs from our testing equipments.

3.2.1 Circuit Diagram

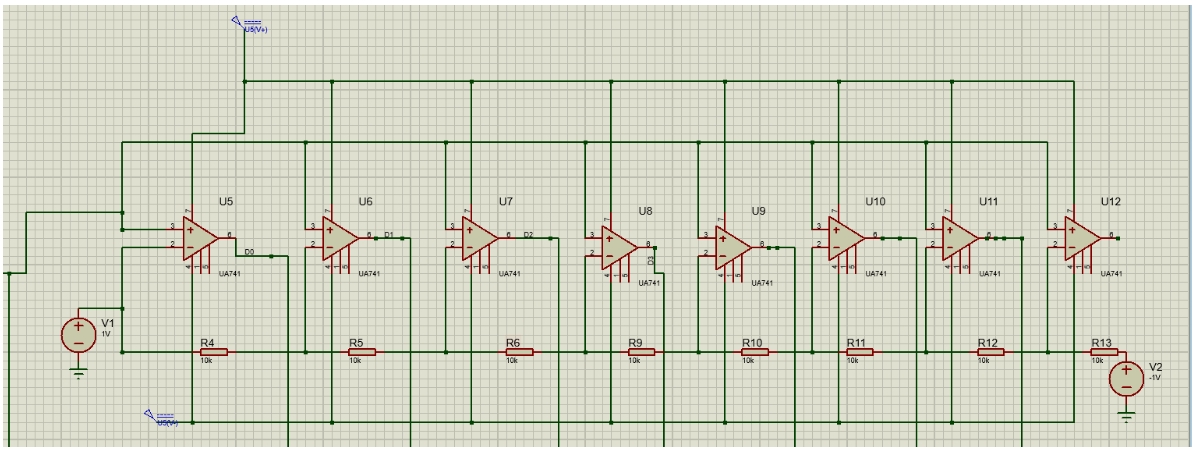
Sampling Circuit using N-Mos:



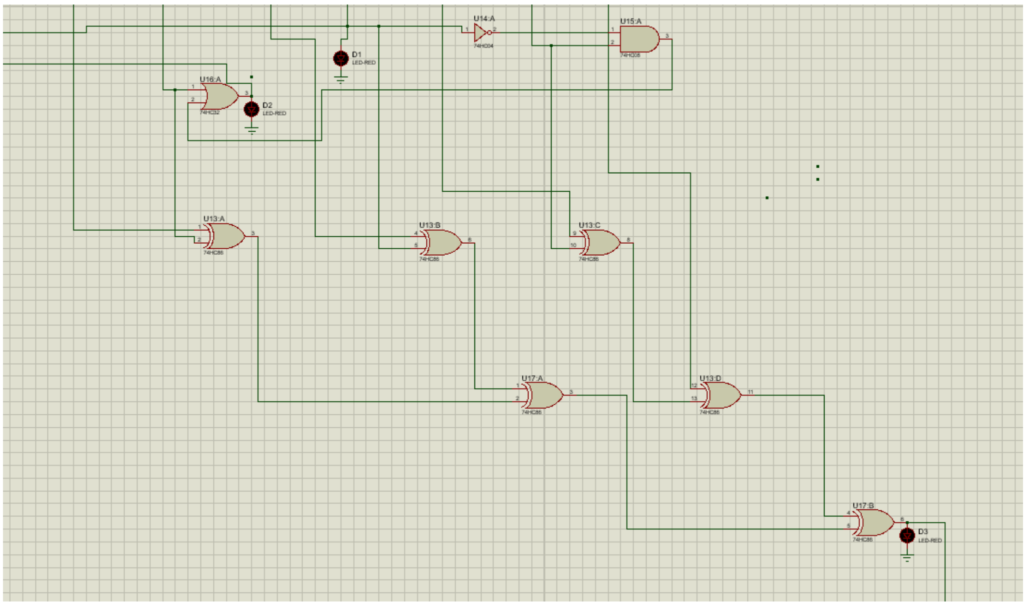
Sampling Circuit using LF398N:



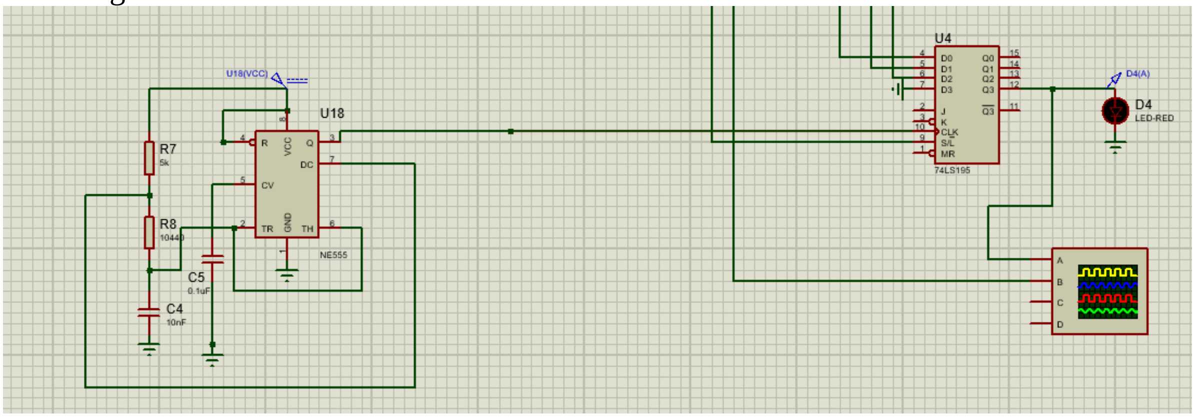
Quantizer Circuit:



Encoder Circuit:

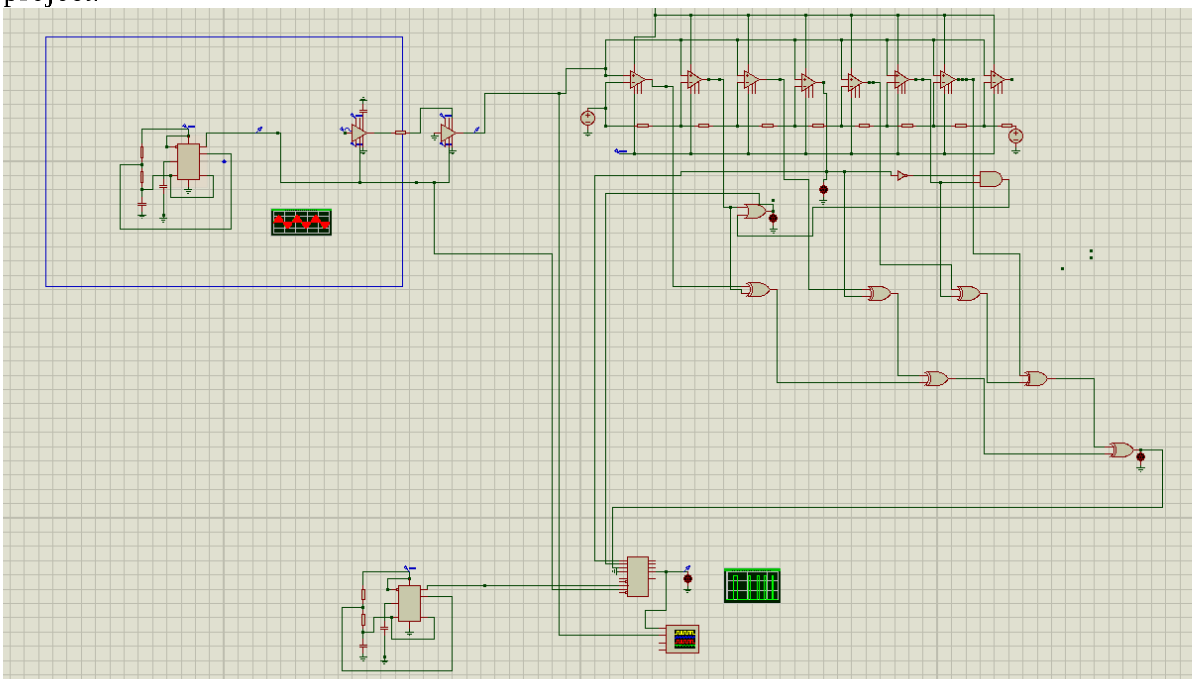


Shift Register Circuit:



3.3 Simulation Model

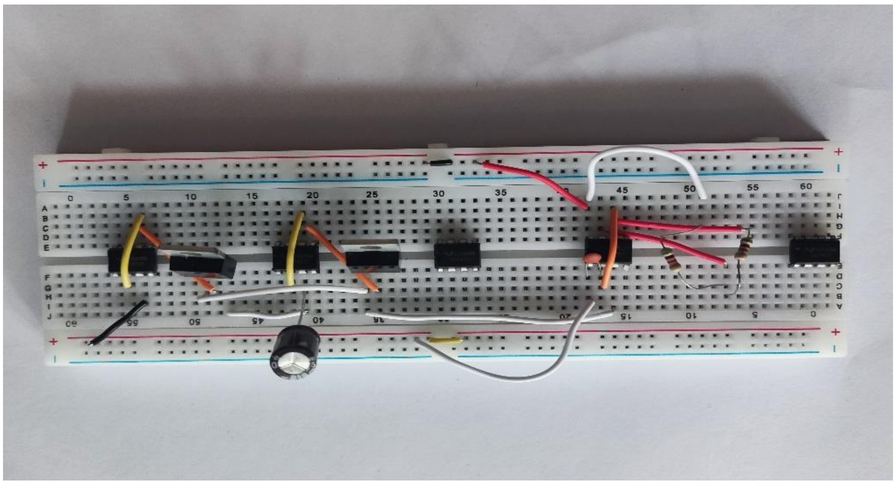
The overall Simulation was first done in LTspice but LtSpice has no logic gates into it but we needed logic gates for building the encoder. So we moved to Proteus to simulate the project.



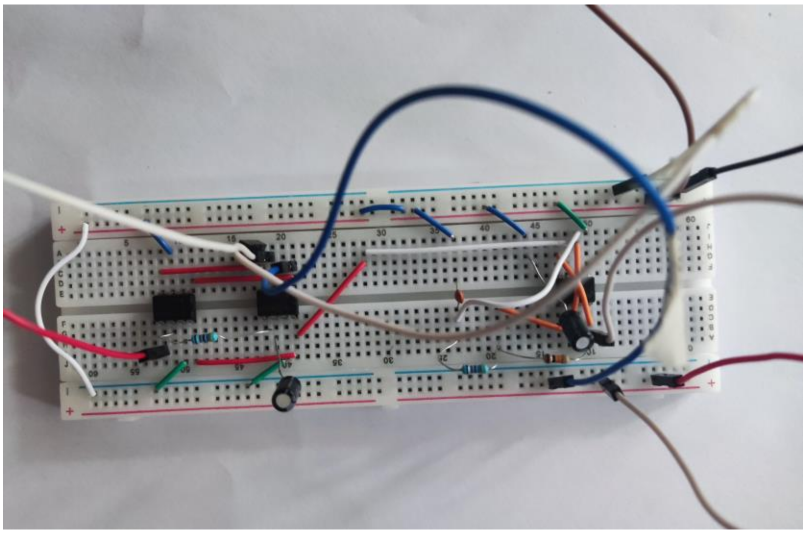
4 Implementation

4.1 Description

Sampler and Hold Circuit: The signal first goes through a buffer amplifier to lessen the impact of loading. Next, it encounters an NMOSFET that toggles on and off using a pulse, serving as the sampling signal. This switched output then travels through a capacitor (with a value of '104') to retain the output temporarily before being buffered again to another sampler, which employs the same NMOSFET but lacks a capacitor. Ultimately, it proceeds to the subsequent stage of operation. This circuit configuration is supplied.

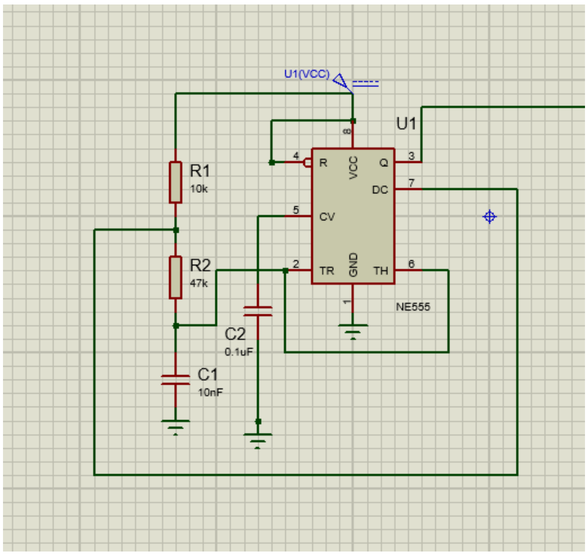


Our sampler which we made using the N-mos didn't work properly ,it didn't show the negative portion of the signal during the sampling process ,So we have used two LF398N for our flat top sampling. We have provided the same pulse to the LF398N and achieved flat top sampling. Our clock also works perfectly.

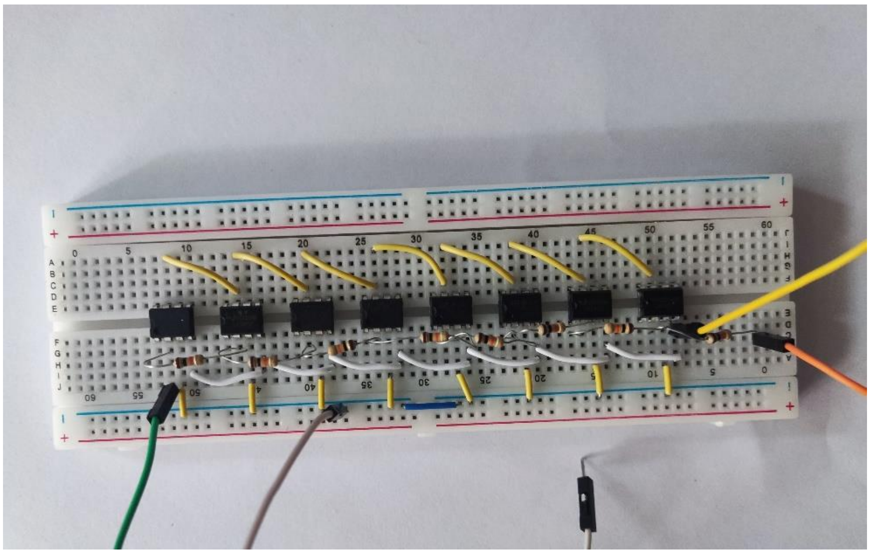


Sampling Pulse:
For sampling pulse , we have used a 555 timer IC in astable mode. PIN-3 is the output (connected to the NMOSFETS), Based on the characteristics of R3, R4 and C3, the up time and down time of the pulse are:
$T_{on} = \ln(2)*(R1+R2)*C1;$

$T_{off} = \ln(2)*R2*C1$;



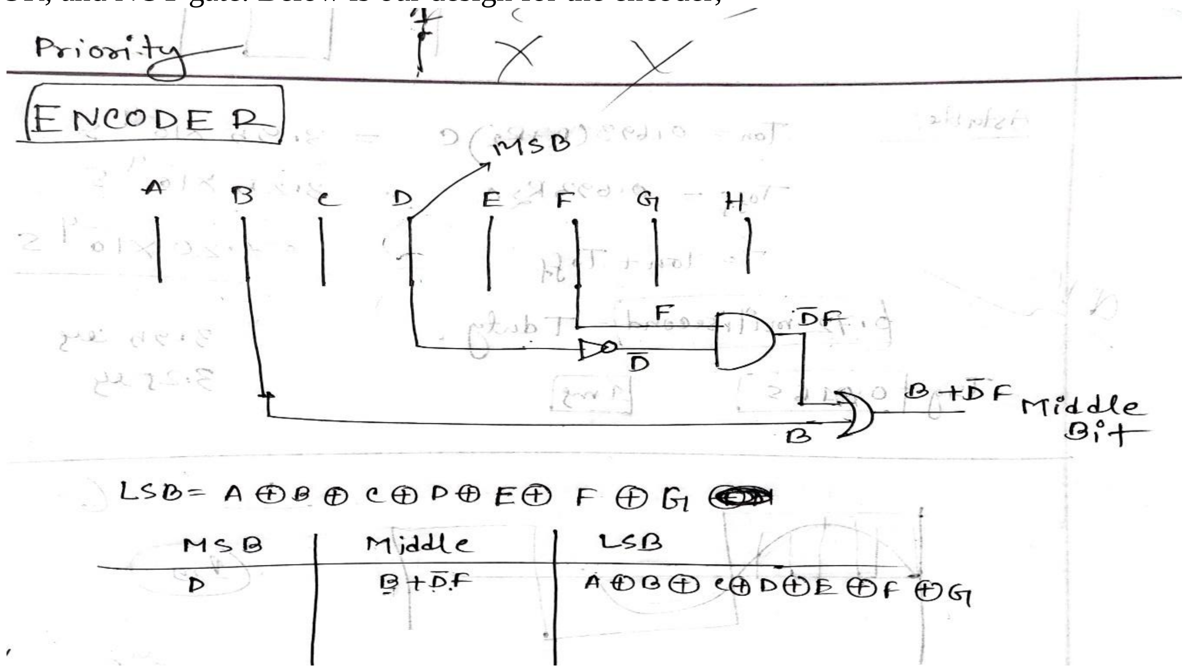
Quantizer: The setup comprises eight UA741-based comparators and eight resistors arranged in series to establish quantization levels. Our aim was to build a 3-bit quantizer. Each PIN-2 of the comparators is linked to one end of each resistor. As the resistors span between two distinct voltage sources, +Vcc1 and GND (where -Vcc2 is set as GND for simplicity), eight progressively increasing voltage values are fed into the PIN-2s. We designate +Vcc1 as 5.0 V.



We have biased the op amp by our DC battery to +9V and the negative bias of the op amp was provided a little less than the zero voltage. As our op amps are not ideal op amps, there will be some voltage in the 6 no pin even if the inverting input is greater than the non inverting input. As we are doing a 3-bit Quantizer ,each level has $5/2^3$=8 Voltage levels. This voltage levels can be changed according to our need. By testing with a fixed dc voltage level we verified that our quantizer works precisely.

Encoder: We have built a 3-bit Encoder from scratch. We didn't use any IC for encoding. We have designed the priority encoder using different logic gates including AND, OR, X-

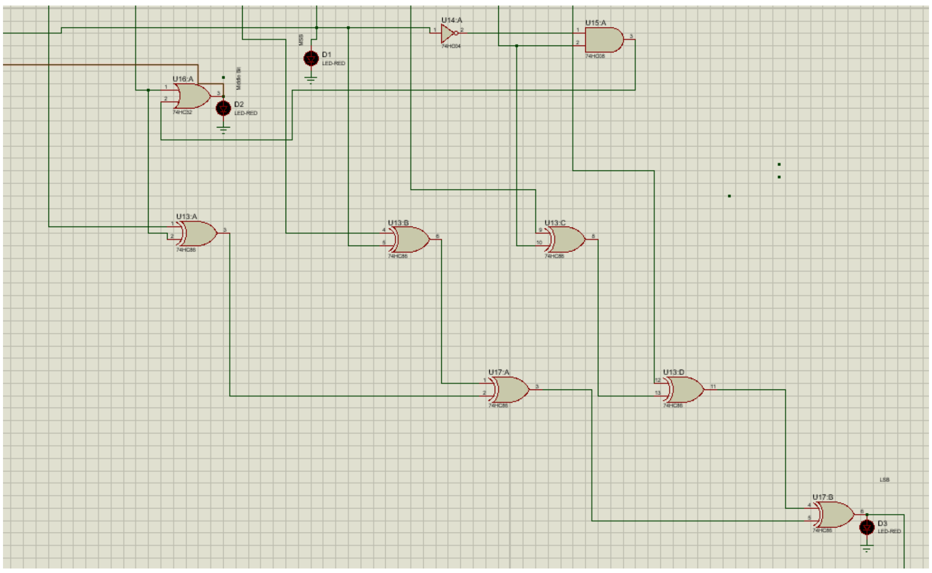
OR, and NOT gate. Below is our design for the encoder,



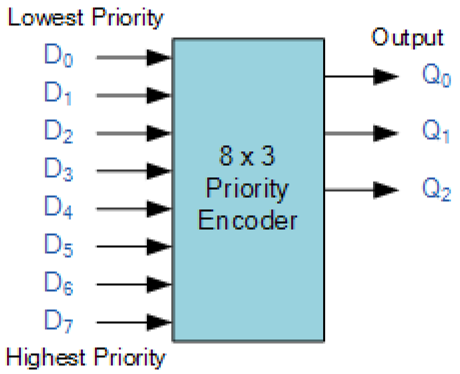
We have used 1 And, Or, Not gate and 2 X-or gates to implement the Encoder. The model number of the gates are

1. SN74HC86N (X-Or gate)
2. SN74HC04N (Not Gate)
3. SN74HC08N (And Gate)
4. SN74HC32N (Or Gate)

The biasing voltage for the gates are 2-5V range for safe operation.



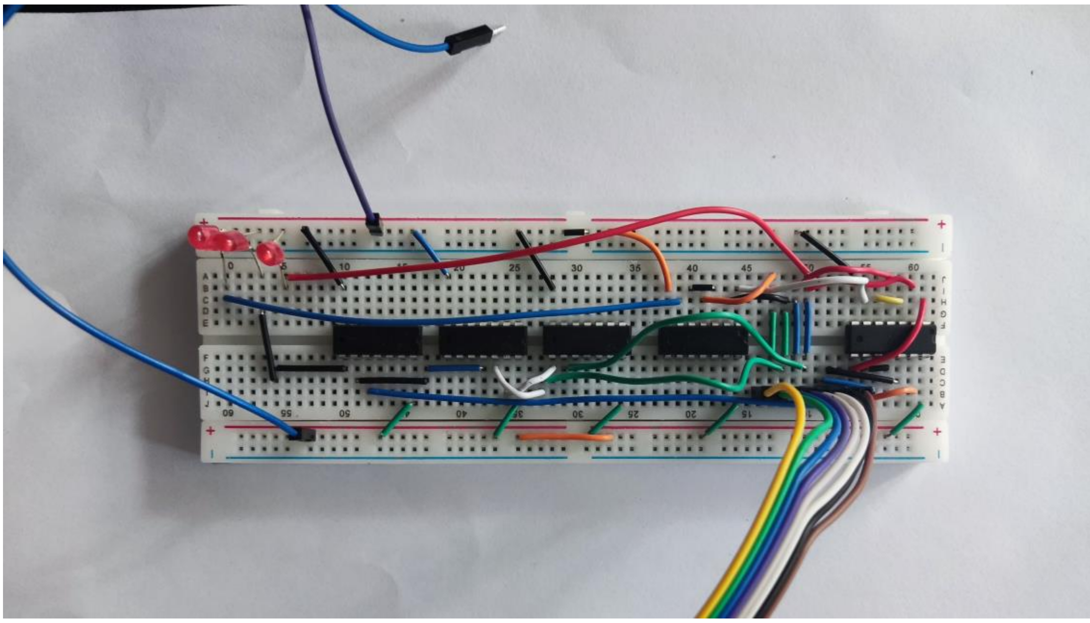
The encoder will provide the equivalent binary bits for a quantization level. The truth table for the priority encoder is given below.



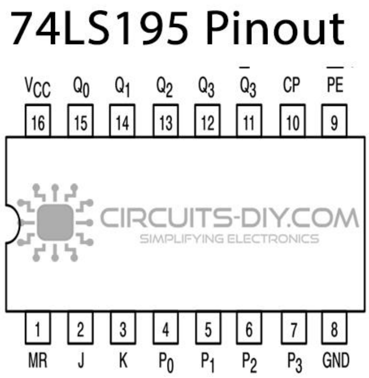
Inputs								Outputs		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

X = dont care

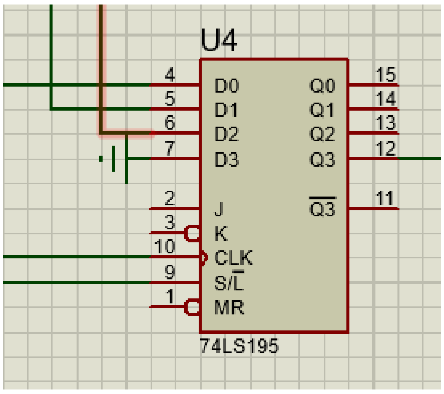
The function of the encoder is evident from the truth table provided earlier. It functions as a priority encoder, with input 7 holding the highest priority. The output of the priority encoder corresponds to the active input with the highest priority. Consequently, when a higher-priority input is detected, all lower-priority inputs are disregarded. The hardware implemented circuit is shown below,



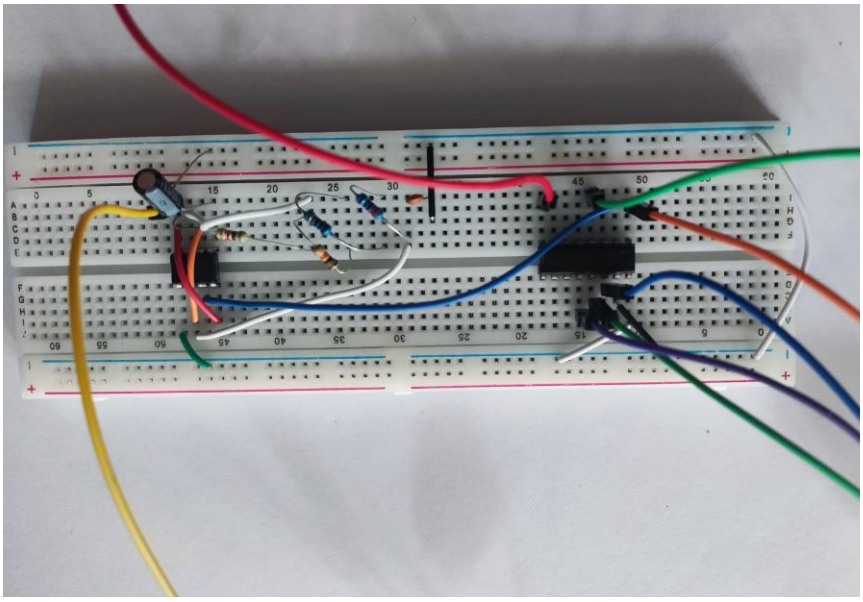
Shift Register: The encoder's output initially provided parallel data. However, for our project's needs, we sought a sequential data sequence. Additionally, we aimed to temporarily store the data and transmit them sequentially through the channel in a controlled manner, synchronized by clock pulses. To achieve this, we opted for a 3-bit Parallel In - Series Out (PISO) shift register.



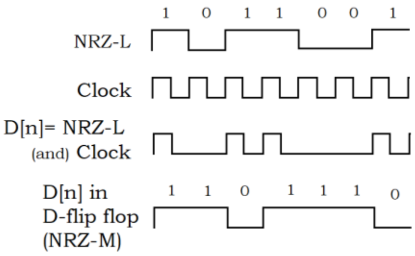
In our implementation, we utilized a 4-bit 74LS195AP shift register. This register features 4 input and 4 output bits. We configured it for serial input and parallel output operation. Three input pins were linked to the encoder's output, while one pin was grounded, as we were utilizing 3-bit encoding. Consequently, the Most Significant Bit (MSB) of the shift register's output was redundant.



It takes two synchronizing clock pulse in its 9 and 10 no pin. The input pulse for 9 no pin is actually the same clock that we used earlier for smapling. The pulse for input 10 is the pulse that we designed. In reality, achieving precise synchronization between the load pulses and the clock pulses was challenging. Both the load pulses and the clock pulses were produced by separate timer circuits. Considering each sample consisted of 3 essential bits and 1 redundant bit, totaling 4 bits to be shifted with every sampling pulse, the clock frequency was precisely set to be 4 times the sampling frequency. To transfer the inputs to the output, a clock pulse had to coincide with the load pulse, followed by 3 additional clock pulses. Hence, meticulous adjustment of the timer circuits was necessary to achieve the intended output.The output of the shift register is NRZ-L.



Other Line Codes: Most important parts of PCM is done. The output of shift register that is out NRZ-L along with a clock pulse of 50% duty cycle can be feed through an AND Gate (IC-7408). Referencing the gate we used earlier building our encoder, the resulting output is then passed through a D-flip-flop. The transitions from 0 to 1 will appropriately trigger the flip-flop, yielding the final NRZ-M line code. In the NRZ-L line code, a transition occurs whenever there is a 1. Considering the initial reference word as 0, the output will resemble the 4th line code. But our flip flop didn't work.



For NRZ-S, By inverting the NRZ-L using an OR Gate (IC-7404) used earlier in making of the encoder and subject it to the same process as for NRZ-M. Consequently, the flip-flop identifies transitions from 0 to 1 where there are actually transitions from 1 to 0, fulfilling our primary objective.
But Our flip-flop IC didn't work at all. Also due to out short time for testing we couldn't configure a way to bias all the circuits and boards. So we couldn't see the later line codes. But our sampler,Quantizer, Encoder, Shift register works perfectly.

5 Design Analysis and Evaluation

5.1 Novelty

Sampling circuit: Generally in sampling nmos is used for sampling and hold circuit formation.But we have used a different technique by op amps to make the sampling circuit.

5.2 Design Considerations

5.2.1 Considerations to public health and safety

The theoretical part behind the PCM process is to make communication electronic hardware dependent. The design we are working with, does not contain any kind of elements that can be a matter of concern for a common person.

In some worst cases, some short-circuitry can happen, but we designed our apparatus very low voltage dependent which will not be able to hamper a lot or will not be fatal.

5.2.2 Considerations to environment

The hardware we used are made up of plastic and different materials which are not eco-friendly.
But those can be used for a long time and can help us lessening our effort in making communication easier.So the little impact on environment can be overlooked.

5.2.3 Considerations to cultural and societal needs

Society is created on the basis of communication. So communication is needed. But to make it easier we can use PCM in a lot of places.It can have a good impact in building strong society and good culture.

5.3 Investigations

5.3.1 Literature Review

5.3.2 Experiment Design

Experiments were set up to check if our project has a favorable result or not.So we made some experiment like-
1. Sampling circuit characteristics monitoring
2. Quantization level detection
3. Comparator output monitoring
4. Encoder output characteristics of AND, OR, XOR gate monitoring
5. LED blink to code conversion and matching with the input message.

5.3.3 Data Analysis and Interpretation

After analysis , we found our circuit to work well for PCM process.

5.4 Limitations of Tools

- **Voltage source common grounding problem:** We were finding it difficult to make a common ground for a lot of voltage sources. the common ground problem was making some voltage level very high which were not appropriate for our work.

- **Biasing of op amp problem:** We faced some problems in making the op amp output 0. If we were feeding 0 volt in the negative bias voltage lead, we were having output of a voltage level larger enough to distort the encoder operation.

- **Quantization level determining problem:** The resistances we used were not ideal.

So we had problem in exactly pinpoint different quantization level.

- **Encoder biasing problem:** Encoder was to bias with a voltage level that can give binary output with respect to input. But such a voltage level was not easy to maintain.

- **XOR-OR gate garbage output problem:** In XOR and OR gate IC, for completely 0 voltage inputs we were getting some positive outputs which were not ideally good to go.

6. Reflection on Individual and Teamwork

6.1: Individual Contribution of Each Member

Serial No.	Description	IDs
1	Research and resource collection	All
2	Theory	All
3	SImulation	2006061,2006064
4	Sampler circuit design	2006062,1706110
5	Quantizer circuit design	2006061,2006065
6	Encoder circuit design	2006064,2006060
7	Shift register circuit design	2006064,2006061
8	Testing Phase 1	2006062,2006064, 2006065,1706110
9	Testing Phase II	2006060,2006061, 2006064,2006065

6.2: Mode of Teamwork

The modes of teamwork we used are:

I. **Collaborative Teamwork:** We used this mode the most. Most of the work related to our project was conducted through collaborative teamwork. Often, we used our hall room to get together and work on the project. During this mode, we shared different opinions on how we could easily achieve something, we discussed important topics related to the project and shared our responsibilities.

II. **Divided Responsibilities:** In this mode, team members were assigned specific tasks. Coding, circuits design etc. were assigned to each member of the group. This way the project has been done before due date.

III. **Leader-Driven Teamwork:** To make decisions fast, get direction, delegate tasks, we used one team leader. As some are attached and residents are from different Hall, one team leader was necessary to guide all of us. And used the Hall room of our team leader. Several necessary decisions were taken quickly by our team leader.

IV. **Virtual Collaboration:** Although we used to go to the hall room for project, we used virtual platform to see the project advancement whenever there was a necessity. It gave a linkage between us through different hall and resident.

6.3: Log Book of Project Implementation

Description	Start Date	End Date
Initial Design	January 22, 2024	January 24, 2024
Buying Parts	February 1, 2024	February 1, 2024
Circuits Implementation by Coding and LTspice and Proteus	February 7, 2024	February 10, 2024
Circuits Implementation on Breadboard	February 14, 2024	February 17, 2024
Testing Phase I	February 21, 2024	February 23, 2024
Testing Phase II	February 24, 2024	February 27, 2024

7. Executive Summary

The primary goal of PCM line code generation is to convert analog signals into a digital format for efficient transmission and subsequent reconstruction at the receiving end. This process involves the representation of analog signals as a sequence of discrete pulses, where the accuracy and efficiency of encoding are paramount. The principles which need to be followed for this purpose are:

Sampling: The analog signal is sampled at regular intervals to capture its amplitude at discrete points in time. This forms the basis for subsequent digital representation.

Quantization: The sampled values are quantized, meaning they are assigned discrete numerical values to represent the amplitude. The precision of quantization determines the fidelity of the digital representation.

Encoding: The quantized values are then encoded into a binary format using line codes. Common line codes include Non-Return-to-Zero (NRZ), Return-to-Zero (RZ), and Manchester. The choice of line code influences factors such as bandwidth efficiency and error detection capabilities.

But the challenges that may arrive are:

Bandwidth Efficiency: Selecting an appropriate line code is crucial to optimize bandwidth usage. Techniques such as Differential Pulse Code Modulation (DPCM) can be employed to reduce the number of bits required for encoding without sacrificing signal quality.

Error Detection and Correction: Incorporating error-detection mechanisms within the line code, such as parity bits or cyclic redundancy checks, enhances the robustness of the communication system, allowing for the identification and correction of transmission errors.

In summary, Pulse Code Modulation Line Code Generation is a fundamental process in digital communication, converting analog signals into a digital format for efficient transmission. The selection of appropriate line codes is vital in achieving optimal bandwidth efficiency, error detection, and signal fidelity. Advancements in line code generation contribute significantly to the reliability and performance of modern communication systems.

Caution
When engaging in the PCM line code generation project, one should be careful on the following things:
Input Signal Sensitivity: To take note of the input analog signal's properties. Unpredictable or extreme input signals have the potential to cause inaccurate digitization and negatively affect the digital output. Should verify that the input signal complies with the given specifications.
Configurable Parameters: When modifying configurable parameters, such as quantization levels, sampling rates, and line coding techniques, one should proceed with care. Inadequate configurations may lead to information loss, signal distortion, or less than ideal performance.
Real-time Process Monitoring: One should keep an attentive eye on the output and system logs during real-time signal processing to quickly spot any unusual behaviors or problems. Also keep an eye out for any alerts or cautions that could point to problems with the signal processing phases.

8. Project Management and Cost Analysis

a. Bill of Materials

Materials	Price (BDT)
Bread Board	500
UA-741	225
Resistors	50
NE 555 Timer	50
Capacitors	100
7408 Quad And Gate	30
7432 Quad Or Gate	30
LED	40
74HC86N IC Gate XOR	60
7404 Hex Inverter IC	30

LF398 Sample and Hold Amplifier IC	300
4 Bit Parallel access Shift Resistor IC	40
IRF540 N Channel MOSFET	165
Jumper Wires	250
Total	**1870**

9. Future Work

This project acts as the foundation of a lot of potential work that can be done. Some of them are-

- We implemented NRZ-L line coding but there are room for other line coding techniques like NRZ-M, Manchester, Differential Manchester etc.
- We used a 3-bit Quantizer for this project which can be improved by implementing higher bit Quantizer to get more quantization levels.
- Use of better parts would make the project more efficient and accurate.
- As we only worked with Pulse Code Modulation, this project can be extended to Pulse Code Demodulation.

10. References

1. Lathi, B. P., & Zhi Ding. (2010). Modern digital and analog communication systems. Oxford University Press.

Chapter 6, Pulse Code Modulation

2. Storr, Wayne. "Priority Encoder and Digital Encoder Tutorial." *Basic Electronics Tutorials*, 1 Aug. 2019, electronics-tutorials.ws/combination/comb_4.html. Accessed 16 Feb. 2024.